



LOB 1003 - Cálculo 1

Lista de exercícios 2 - Parte 2
2º semestre de 2025

1.

a) $\frac{dy}{dx} = \frac{-y^2 - 2xy}{x^2 + 2yx}$

b) $\frac{dy}{dx} = \frac{1 - 2y}{2x + 2y - 1}$

c) $\frac{dy}{dx} = \frac{-2x^3 + 3x^2y - xy^2 + x}{y - x^3 + x^2y}$

d) $\frac{dy}{dx} = \frac{1}{y} \frac{1}{(x+1)^2}$

e) $\frac{dy}{dx} = -\frac{(1 - \sqrt{x})}{\sqrt{x}}$

f) $\frac{dy}{dx} = \cos^2 y$

g) $\frac{dy}{dx} = \frac{2e^{2x}}{3\cos(x+3y)} - \frac{1}{3}$

h) $\frac{dy}{dx} = \frac{-y^2}{y \sin(1/y) + yx - \cos(1/y)}$

i) $\frac{dy}{dx} = \frac{-y}{x}$

2.

a) $y' = \frac{-x}{y}, \quad y'' = -\frac{1}{y^3}$

b) $y' = \frac{xe^{x^2} + 1}{y}, \quad y'' = \frac{(y^2 + 2x^2y^2 - 2x)e^{x^2} - 1 - x^2e^{2x^2}}{y^3}$

c) $y' = \frac{\sqrt{y}}{1 + \sqrt{y}}, \quad y'' = \frac{1}{2(1 + \sqrt{y})^3}$

d) $y' = \frac{-x^2}{y^2}, \quad y'' = \frac{-2xy^3 - 2x^4}{y^5}$

3.

$$\text{a) } y' = \frac{\sqrt{x(x+1)}}{2} \left(\frac{1}{x} + \frac{1}{x+1} \right)$$

$$\text{b) } y' = \frac{1}{2} \sqrt{\frac{t}{t+1}} \left(\frac{1}{t} - \frac{1}{t+1} \right)$$

$$\text{c) } y' = \sqrt{\theta+3} \operatorname{sen} \theta \left(\frac{1}{2(\theta+3)} + \operatorname{cotg} \theta \right)$$

$$\text{d) } y' = 3t^2 + 6t + 3$$

$$\text{e) } y' = \left(\frac{\theta+5}{\theta \cos \theta} \right) \left(\frac{1}{\theta+5} - \frac{1}{\theta} + \operatorname{tg} \theta \right)$$

$$\text{f) } y' = \frac{x\sqrt{x^2+1}}{(x+1)^{2/3}} \left(\frac{1}{x} + \frac{x}{x^2+1} - \frac{2}{3(x+1)} \right)$$

$$\text{g) } y' = \frac{1}{3} \sqrt[3]{\frac{x(x-2)}{x^2+1}} \left(\frac{1}{x} + \frac{1}{x-2} - \frac{2x}{x^2+1} \right)$$

4.

$$\text{a) } \frac{\sqrt{2}}{2}$$

$$\text{b) } -\frac{\sqrt{3}}{3}$$

$$\text{c) } \frac{4+\sqrt{3}}{2\sqrt{3}}$$

$$\text{d) } 1$$

$$\text{e) } -\sqrt{2}$$

5.

$$\text{a) } y' = \frac{-2x}{\sqrt{1+x^4}}$$

$$\text{b) } y' = \sqrt{\frac{2}{1-2t^2}}$$

$$\text{c) } y' = \frac{s}{|2s+1|\sqrt{s^2+s}}$$

$$\text{d) } y' = \frac{-2}{|x^2+1|\sqrt{x^2+2}}$$

$$\text{e) } y' = \frac{-1}{\sqrt{1-t^2}}$$

$$\text{f) } y' = \frac{-1}{2\sqrt{t}(1+t)}$$

$$\text{g) } y' = \frac{1}{\operatorname{tg}^{-1}x} \frac{1}{1+x^2}$$

$$\text{h) } y' = \frac{-1}{\sqrt{e^{2t}-1}}$$

$$\text{i) } y' = \frac{-2s^2}{\sqrt{1-s^2}}$$

j) $y' = 0$

k) $y' = \operatorname{sen}^{-1} x$

8.

a) $f'(x) = x \cosh x$

b) $h'(x) = \operatorname{tgh} x$

c) $y' = 3e^{\cosh 3x} \sinh 3x$

d) $f'(t) = -2e^t \operatorname{sech}^2 e^t \operatorname{tgh} e^t$

e) $G'(x) = \frac{2 \operatorname{cosech} x \cotgh x}{(1 + \operatorname{cosech} x)^2}$

f) $y' = \frac{1}{\sqrt{x-1}} \frac{1}{2\sqrt{x}}$

g) $y' = \sinh^{-1} \left(\frac{x}{3} \right)$

h) $y' = -\operatorname{cosec} x$

9. $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}.$

10.

a) $\frac{dV}{dt} = \pi r^2 \frac{dh}{dt}$

b) $\frac{dV}{dt} = 2\pi h r \frac{dr}{dt}$

c) $\frac{dV}{dt} = \pi r^2 \frac{dh}{dt} + 2\pi h r \frac{dr}{dt}$

11.

a) 1 volts/s

b) -1/3 A/s

c) $\frac{dR}{dt} = \frac{1}{I} \left(\frac{dV}{dt} - \frac{V}{I} \frac{dI}{dt} \right)$

d) 3/2 ohms/s

12.

a) $\frac{ds}{dt} = \frac{x}{\sqrt{x^2 + y^2}} \frac{dx}{dt}$

b) $\frac{ds}{dt} = \frac{1}{\sqrt{x^2 + y^2}} \left(x \frac{dx}{dt} + y \frac{dy}{dt} \right)$

c) $\frac{dx}{dt} = \frac{-y}{x} \frac{dy}{dt}$

13.

a) $L(x) = 10x - 13$

b) $L(x) = 2$

c) $L(x) = x - \pi$

d) $L(x) = 1 + kx$

14.

a) $dy = \left(3x^2 - \frac{3}{2\sqrt{x}} \right) dx$

b) $dy = \left(\frac{2 - 2x^2}{(1 + x^2)^2} \right) dx$

c) $dy = \left(\frac{1 - y}{3\sqrt{y} + x} \right) dx$

d) $dy = \frac{5}{2\sqrt{x}} dx$

e) $dy = 4x^2 \sec^2 \left(\frac{x^3}{3} \right) dx$

f) $dy = 2(x \sin 2x + x^2 \cos 2x) dx$

g) $dy = \frac{1}{2\sqrt{t}} \sec^2 \sqrt{t} dt$

h) $dy = \frac{3 \operatorname{cosec} (1 - 2\sqrt{x}) \cotg (1 - 2\sqrt{x})}{\sqrt{x}} dx$

i) $dy = \frac{e^{\sqrt{x}}}{2\sqrt{x}} dx$

j) $dy = \frac{2x}{1 + x^2} dx$

k) $dy = \frac{2xe^{x^2}}{1 + e^{2x^2}} dx$

l) $dy = \frac{-1}{\sqrt{e^{-2x} - 1}} dx$

m) $dy = \frac{e^{x/10}}{10} dx$

n) $dy = \frac{x}{\sqrt{3 + x^2}} dx$